Functional Requirement Specification, Nathaniel Johnson

Stakeholders

- Moviegoers
- Theater Managers
- Theater Staff
- System Administrators

Primary Actors

- Customer: This actor can browse movies and showtimes, select seating and view their booking confirmation.
- Theater Staff: The staff will help customers with ticket purchases and monitor attendance and seating.

Secondary Actors

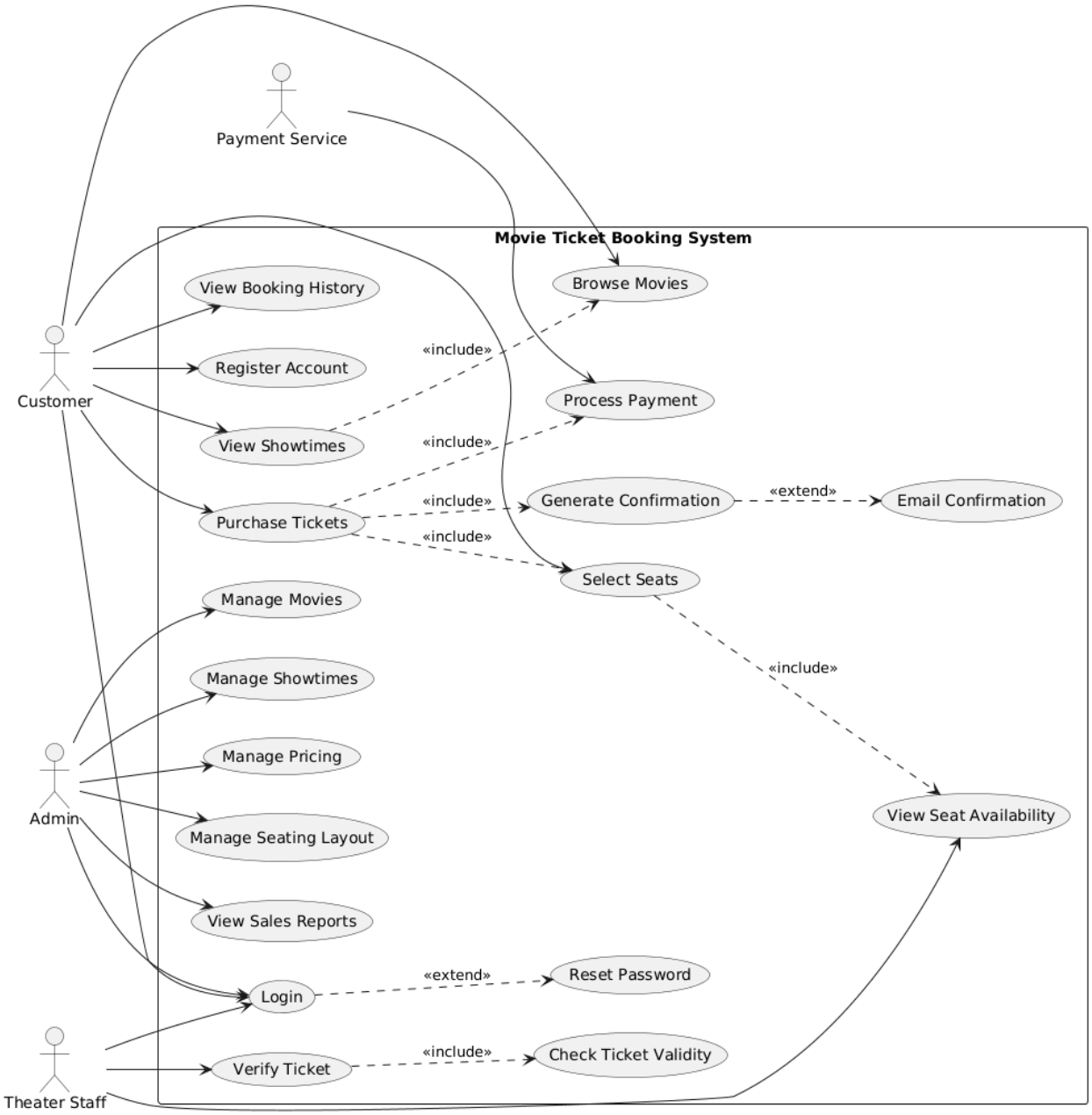
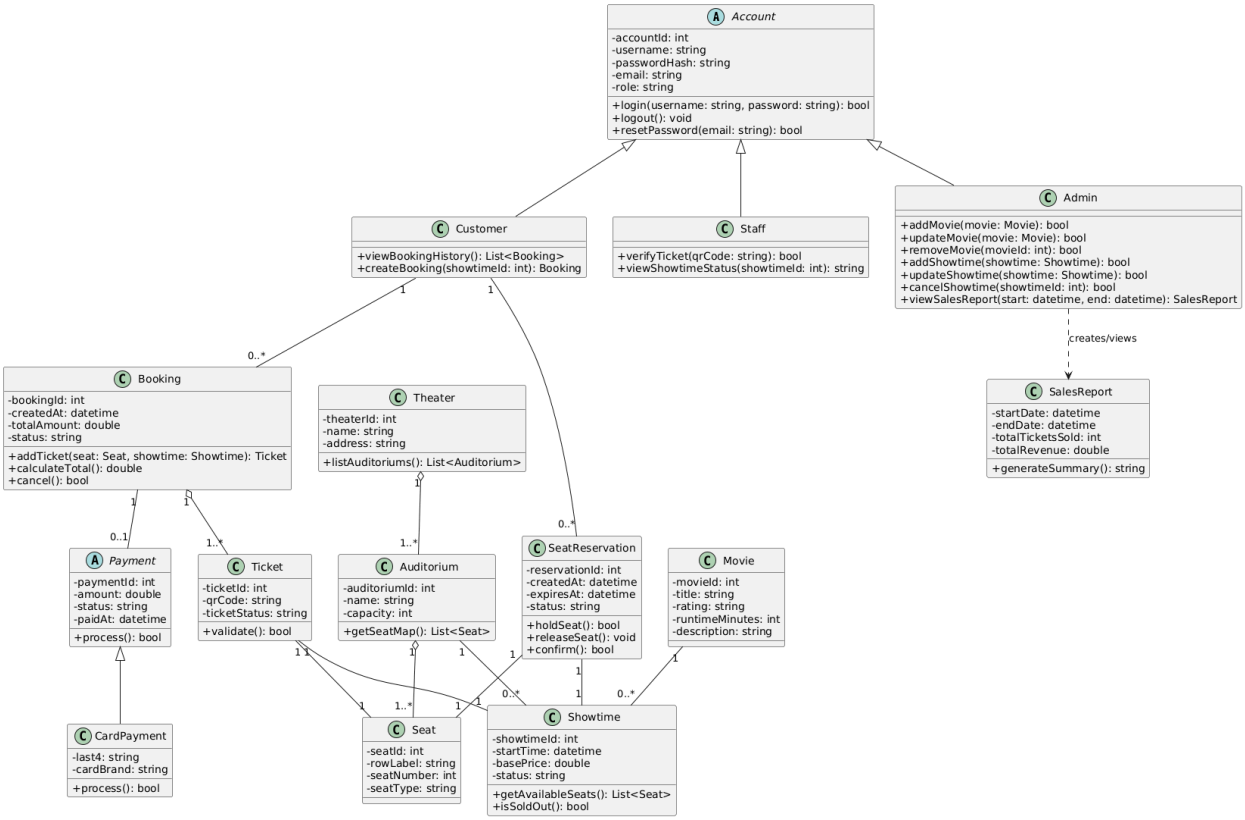
- Admin: They can manage movies, showtimes, seating layouts and prices.
- System: This is responsible for tracking seat availability, prevent double bookings and generate confirmation reports.

Use Cases

- Admin (Total: 18)
    - Add movie: Add a new movie to the system (2)
    - Update movie: Edit or remove movie details (2)
    - Add showtime: Create showtimes for movies (2)
    - Modify pricing: Adjust ticket prices (4)
    - Manage seating layout: Configure seating availability (2)
    - Login/Logout: Access admin features securely (4)
    - View sales report: View ticket sales and attendance data (2)
- Customer (Total: 12)
    - Browse movies: View current and upcoming movies (2)
    - View showtimes: See available showtimes for a movie (2)
    - Select seats: Choose available seats (2)

    - Purchase ticket: Complete ticket purchase (2)
    - Receive confirmation: Get booking confirmation (2)
    - View booking history: Review past purchases (2)
- Theater Staff (Total: 18)
    - Login/Logout: Access staff tools (2)
    - View showtimes: Check schedules (2)
    - Assist purchase: Help customers complete ticket purchase (2)
    - View seat availability: Monitor seating in real time (2)
    - Verify ticket: Confirm valid ticket at entry (2)
    - View attendance: Monitor audience size (2)
    - Update account: Manage profile details (2)
    - View account: Review account information (2)
    - Handle issues: Assist with seating or booking problems (2)
- System (Total: 9)
    - Assign seats: Reserve selected seats for a customer (2)
    - Prevent double-booking: Ensure seats are not sold twice (3)
    - Show sold-out status: Indicate when a showtime is full (2)
    - Update availability: Display real-time seat availability (2)

Use Case Diagram



Class Diagram